

Luca Lombardo

MATHEMATICS · COMPUTER SCIENCE

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Education

BSc in Mathematics

UNIVERSITY OF PISA (THESIS: EFFICIENT SUCCINCT DATA STRUCTURES ON DAGs)

University of Pisa, Italy

9 May 2025

MSc in Computer Science

UNIVERSITY OF PISA (THESIS: COMPRESSED REPRESENTATION OF SET COLLECTIONS)

University of Pisa, Italy

Expected: 29 May 2026

Experience

Research Fellowship

ISTITUTO DI SCIENZA E TECNOLOGIE DELL'INFORMAZIONE "A. FAEDO" (CNR-ISTI)

Pisa, Italy

Feb 2026 - Aug 2026

- Finite elements methods software for thermo-mechanical analysis and associated graphical interfaces
- Implementing a Rust graphical interface for a Fortran finite element solver

System Administrator

UNIVERSITY OF PISA, DEPARTMENT OF MATHEMATICS, PHC

Pisa, Italy

January 2024 - Present

- Created and maintain servers and services for the department of mathematics
- SysOps tasks and recently also DevOps tasks, such as setting up CI/CD pipelines for the department's open-source projects.

Teaching Assistant: Algorithms and Programming

UNIVERSITY OF PISA, DEPARTMENT OF COMPUTER SCIENCE

Pisa, Italy

Feb 2026 - Apr 2026

- Tutor for the Algorithms and Programming course in the Department of Computer Science.
- Mentored first-year students on the basis of Algorithms and Data Structures

Teaching Assistant: Parallel Computing from the Infrastructure to the Mathematics

UNIVERSITY OF PISA, DEPARTMENT OF MATHEMATICS

Pisa, Italy

April 2023 - June 2023

- Assembled, configured and maintained a 20-node HPC cluster for course use
- Mentored students on parallel algorithms (C/C++/Julia, OpenMPI) and parallel environment setup (Linux, Slurm)

Content Editor & Writer

MORETHANTECH

Italy (Remote)

June 2020 - July 2022

- Wrote more than 50 articles on open-source software, learning Linux, fixing Windows and mechanical keyboards.
- Managed a small team of writers, providing feedback and editing to ensure high-quality content.

Publications

High-Throughput GPU-Accelerated Succinct Data Structures

RUST, GPU, SPIR-V, SUCCINCT DATA STRUCTURES

Preprint forthcoming. State-of-the-art GPU-accelerated implementation that outperforms the best existing succinct data structure libraries. Vendor-neutral kernels written in Rust and compiled to SPIR-V, exposed through a plug-and-play API.

Evaluating Rust for Sparse Matrix Computations in Scientific Applications

RUST, SPARSE LINEAR ALGEBRA, HPC, BENCHMARKING

Preprint forthcoming. Comparative evaluation of **Rust** against MKL, PETSc, Eigen, and PSBLAS for sparse matrix computations. Designed an optimized native Rust CSR kernel achieving competitive performance with established C/C++/Fortran libraries.

Projects

[1] Compressed Integer Vector Library

RUST, BIT MANIPULATION, SUCCINCT DATA STRUCTURES

A **Rust library** that provides space-efficient, in-memory representations for integer vectors.

[2] A Cache Efficient, Low Memory, Lanczos Algorithm

RUST, NUMERICAL LINEAR ALGEBRA, CACHE OPTIMIZATION

Two-pass Lanczos with $O(n)$ memory; exploits **cache locality** to outperform standard $O(nk)$ implementations on large sparse systems.

[3] An exact and fast algorithm for computing top-k closeness centrality

GRAPH ALGORITHMS, MULTI-THREADING, OPTIMIZATION, C++

Multi-threaded C++ algorithm to compute top-k centrality efficiently on very large graphs.

[4] Efficient Succinct Data Structures on Directed Acyclic Graphs

BACHELOR THESIS IN MATHEMATICS

A **novel succinct data structure** for DAGs that efficiently support queries while maintaining space usage below the entropy of the underlying graph.